DAY 04 | PRE-READ

How Apps Talk and Remember

A light read before Day 4. About 15 minutes, no code required.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Why read this?

When you order food on an app, your phone, a company's computers and a database all work together in a second or two. Tomorrow you will see how that works, and you will store data in a modern database called MongoDB. You will also draw your first charts.

1. Pictures make numbers speak

A table of 365 daily sales figures is hard to read. A simple line chart shows the trend instantly. Charts help people decide faster. We will use a Python tool called Matplotlib to draw them.

2. The restaurant model of an app

Restaurant	App world	Job
Customer	Client (phone, browser)	Asks for something
Waiter	API	Carries requests and responses, follows the menu rules
Kitchen	Server	Does the actual work
Pantry	Database	Stores the ingredients, which is the data

The customer never walks into the kitchen. They only talk to the waiter using the menu. In the same way, apps never touch the database directly. They talk to an API that follows agreed rules.

3. Two ways to store information

Some databases store data in neat tables, like a spreadsheet. Others store each record as a flexible **document**, like a filled-in form that can have extra sections when needed. MongoDB is a document database. It is popular because real-world information is often messy and nested.

> **Example:** one customer has two addresses and another has five. A document simply holds a list of addresses, without needing a separate table.

4. Four actions on any data: CRUD

- **Create**: add a new contact to your phone.
- **Read**: search for a contact.

- **Update**: change their number.
- **Delete**: remove the contact.

Almost every app you use is CRUD at its heart.

5. Think about these before class

- Pick an app you used today. What was the client, and what data did it ask for?
- What information would you store about a product in an online shop? Is any of it a list?
- Which chart would you use to show your monthly expenses?

Irasshaimase (いらっしゃいませ): welcome, said to customers in shops and restaurants